Katlego Tsotetsi

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Computer Science Student | Aspiring Intern

Skills Summary

- **Languages:** Python, JavaScript, SQL, Java, HTML/CSS
- **Backend & Cloud:** Node.js, Express, PostgreSQL, AWS (S3, IAM), REST APIs, Vercel, Render
- **Data & Machine Learning:** Pandas, NumPy, Scikit-learn, XGBoost, QGIS, ArcGIS Pro
- **Tools & Design:** Git/GitHub, Figma, UI/UX Wireframing-

Portfolio: https://katlego204.github.io/portfolio/

WORK EXPERIENCE

Quantify Your Future **01/2026 - 02/2026**
QYF Quant Intern **remote**

- Engineered predictive flood risk forecasting models for Absa using environmental data, integrating dam levels and climate lag features to establish risk thresholds.
- Developed transaction fraud detection models utilizing Python, evaluating logistic regression coefficients and precision/recall metrics.
- Implemented an XGBoost classifier to analyze digital banking customer engagement, optimizing targeting strategies based on feature importance and ROC AUC scores.

Pixellot - AI-Automated Sports Video and Analytics **09/2024 - 06/2025**

Outsources monitoring capabilities to broadcast companies. In my time of employment, we were assigned to SuperSport.

Live Broadcast Monitoring Operator • Part-time

- *Responsible for monitoring live sports broadcasts hosted by Supersport to ensure stream quality and uptime*. Detected and escalated over **200+ real-time broadcast issues** (audio desync, buffering, signal loss), reducing average response time by **30%** and minimizing viewer disruption. Collaborated with technical teams and onsite operators to resolve broadcast incidents, directly contributing to **fewer post-event complaints and faster root-cause investigations.**

SpatialEdge **08/2024 - 08/2024**
Hackathon Participant - (2024, 2025)

- Built a cloud cost anomaly detection system using Python and pandas during a 48-hour hackathon sponsored by SpatialEdge. Developed a solution to identify outliers in cloud cost data through statistical analysis and data visualization. Collaborated in a team setting to prepare the data, design a detection approach, and present findings to judges.

SUDS Hackathon - 2025
Hackathon Participant - 2025

- In under 2 days, I built a web app that makes learning feel like gaming. I used the power of peer involvement and AI-integration to create an app where you upload your lecture slides, and it takes a DuoLingo approach, making you learn the concepts step by step by step. You can also add friends and find out whether they are studying as well so the you can study together.
- WON 2nd place

LaunchLab
Entrepreneur Bootcamp Participant - 2025

- Learned about LaunchLab student entrepreneur programs, I learned how to make a good pitch and how to know if an idea is worth working on and so much more.
  Co-developed and presented a validated startup idea
  Recognition for innovation and communication excellence as we
- WON 3rd place

EDUCATION

BSc Computer Science • Stellenbosch University 01/2024 - 12/2027

I am currently a student studying Computer Science at Stellenbosch University

Introduction to Computer Science • CS50 07/2023 - 12/2023

I took CS50 classes on python programming and the CS50 Introduction to Computer Science.

National Senior Certificate • Poelano Secondary School

Physical Sciences (A) Life Sciences (A) Mathematics (B) Economics (B) Tourism (A) English (B) Sesotho (C) Life Orientation (A)

CERTIFICATIONS

AWS Certified Cloud Practitioner (In Progress)
Amazon Web Services

FCF - Certified Cybersecurity Fundamentals
Fortinet

Cisco Security Operations Center
Coursera

Microsoft Python Programming Fundamentals
Coursera

nvidia Introduction to Network
Coursera

QYF Alumni cert
SAGEA

PROJECTS

CampusCorner.co.za | Full-Stack Developer 08/2025 - 09/2026

- Architected and deployed a full-stack student e-commerce web application utilizing Node.js, PostgreSQL, and native JavaScript.
- Engineered a scalable image processing pipeline converting uploads to WebP format in-memory, routing them securely to an AWS S3 bucket.
- Implemented a secure authentication flow using Nodemailer and cryptographic tokens, alongside a custom role-based administrative dashboard for real-time database moderation.

AI-Powered EdTech Web App | SUDS Hackathon (2nd Place) 10/2025 - 10/2025

- Built a gamified, peer-integrated learning platform within 48 hours.
- Integrated AI functionality to parse user-uploaded lecture slides and generate step-by-step, Duolingo-style conceptual breakdowns

Cloud Cost Anomaly Detector | SpatialEdge Hackathon 10/2024 - 10/2024

Developed an anomaly detection system utilizing Python and Pandas to identify and visualize statistical outliers in cloud billing data over a 48-hour sprint.

VOLUNTEERING & LEADERSHIP

Matiepreneur Committee
01/2026 - Present

Digital Design & Visuals Officer
Stellenbosch

A proud member of the Matiepreneur Committee, responsible for planning and hosting entrepreneurship events for student founders. Led outreach initiatives, engaged with sponsors, facilitated networking opportunities, and supported student entrepreneurs in gaining visibility and funding to grow their businesses.

- Utilized **Canva** and **Figma** to produce high-quality marketing materials aligned with brand identity
- Collaborated with committee members to translate event goals into clear, engaging visual content
- Enhanced event visibility and student engagement through consistent and impactful design

Kraaie [Stellenbosch University]
02/2024 - 11/2024

Member, Residence Choir
Stellenbosch

Participated in campus events where we had to give group performances at University Choir Competitions and weekly performances for the neighbouring residences.

Chess Committee Leader
05/2020 - 12/2022

Event Coordinator, Team leader
Vereeniging

Organized events, led team strategy sessions, and promoted critical thinking through chess among peers. We were a team of over 50 students and over half of them I personally taught them chess and gave weekly chess lessons for almost 2 years.